INTERNSHIP REPORT

#semanticClimate Internship

BRIC – National Institute of Plant Genome Research (BRIC-NIPGR)

New Delhi, India

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Programme	B.Sc. Botany (Honours with Research)
Duration	1 June 2026 – 22 July 2026

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Acknowledgements

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I am also thankful to the wider #semanticClimate team for their collaboration, feedback, and the opportunity to be part of an interdisciplinary and forward-looking project.

1. Introduction

I joined #semanticClimate as an intern with the objective of gaining exposure to the application of Artificial Intelligence in climate education and science communication. The internship provided an opportunity to contribute to the development and evaluation of AI-powered educational tools while learning collaborative workflows involving climate science, educational technology, and responsible AI.

Throughout the internship, I worked on creative science communication, chatbot quality assurance, and content development for the Climate Academy.

2. Objectives

The internship has helped me achieve the following objectives:

- To understand the role of AI in climate education.
- To contribute to the development and testing of climate education chatbots.
- To communicate climate concepts through accessible visual media.
- To gain exposure to collaborative software development and interdisciplinary project workflows.
- To explore innovative approaches for increasing engagement with climate literacy tools.

3. Activities Undertaken

3.1 Quality Assurance Testing

Conducted systematic testing of the following chatbots:

- Climate Academy Bot (CAbot)
- ClimateInsight (IPCC-based chatbot)

The testing process involved interacting with the chatbot using diverse user prompts and evaluating its responses from both scientific and educational perspectives.

The objective was not only to identify factual inaccuracies but also to assess whether the chatbot communicated climate concepts in a clear, age-appropriate, and engaging manner.

The QA process broadly included:

1. Scientific Accuracy

Responses were checked against accepted climate science concepts presented in the Climate Academy and IPCC reports.

Attention was given to:

- factual correctness
- consistency
- misinformation or hallucinations
- appropriate explanation of scientific terminology

2. Educational Effectiveness

Responses were evaluated for

- clarity of explanation
- logical flow
- suitability for school students

3. User Experience Evaluation

The chatbot was tested from the perspective of a first-time learner.

This involved observing

- conversation flow
- ease of interaction
- response structure
- readability

4. Prompt Testing

Different styles of prompts were used to understand the chatbot's robustness.

This helped identify situations where responses could be improved through prompt engineering or refinement of the knowledge base.

5. Feedback and Reporting

Observations from testing sessions were discussed during project meetings, communicated to the development team, and reported to the issues repository in github to support improvements in the chatbot's educational performance and usability.

Link to the Cabot: https://cabot-semanticclimate.org/

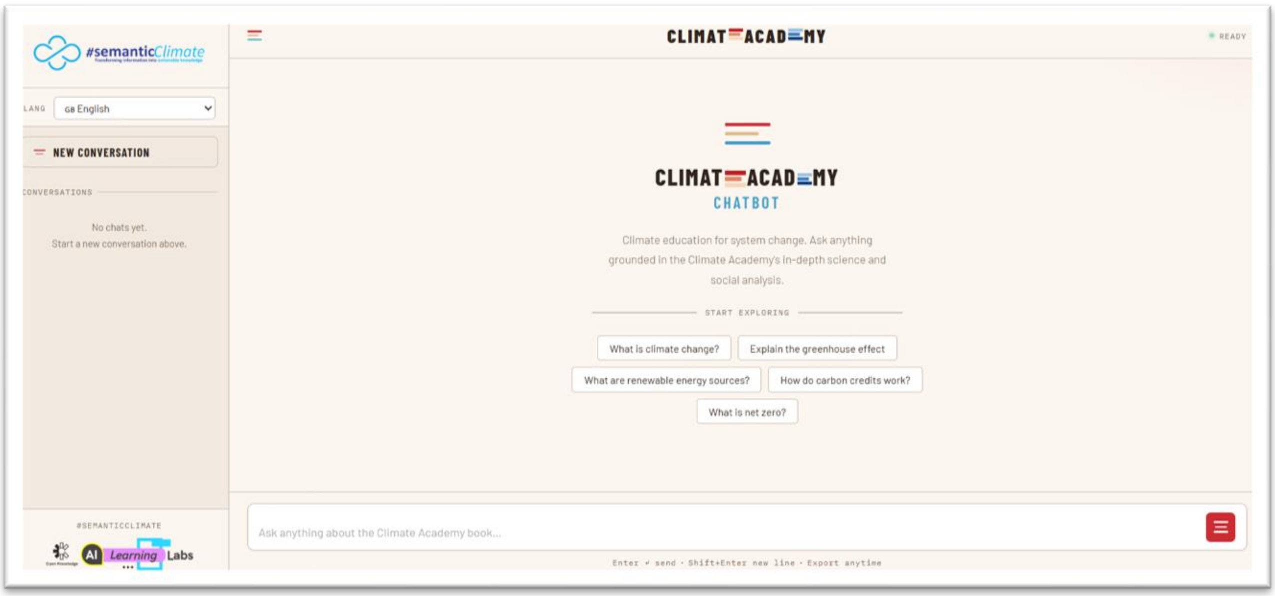
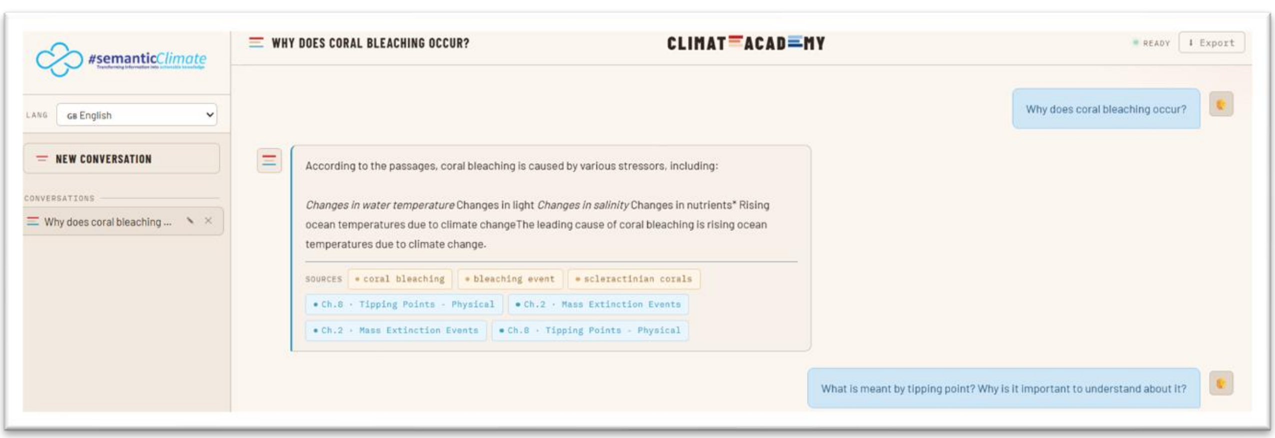
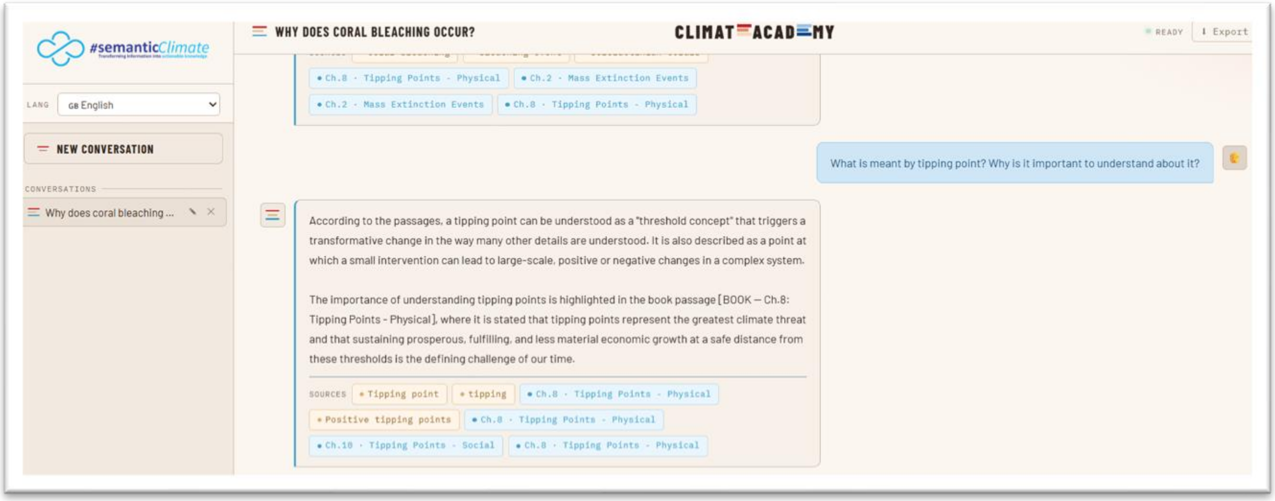


Figure 1: User interface of the Climate Academy chatbot





Figures 2.1, 2.2: Testing conceptual explanations for scientific clarity

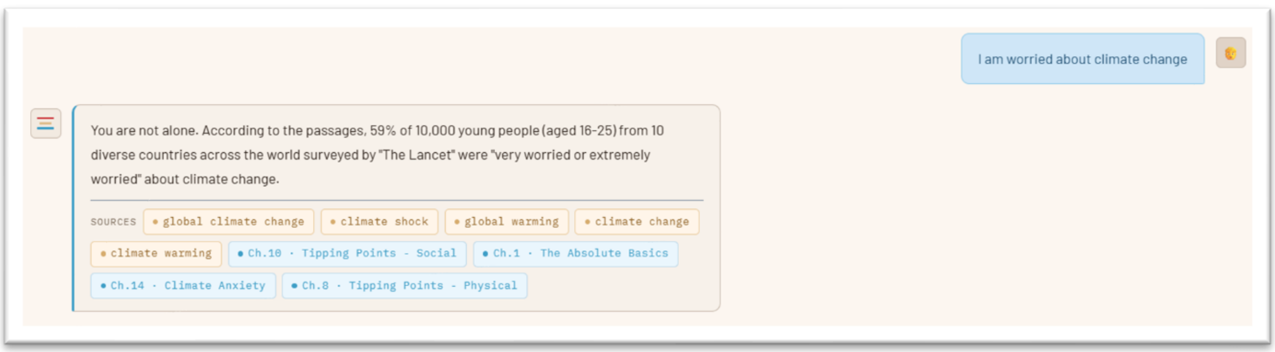


Figure 3: Assessing responses to open-ended learner queries

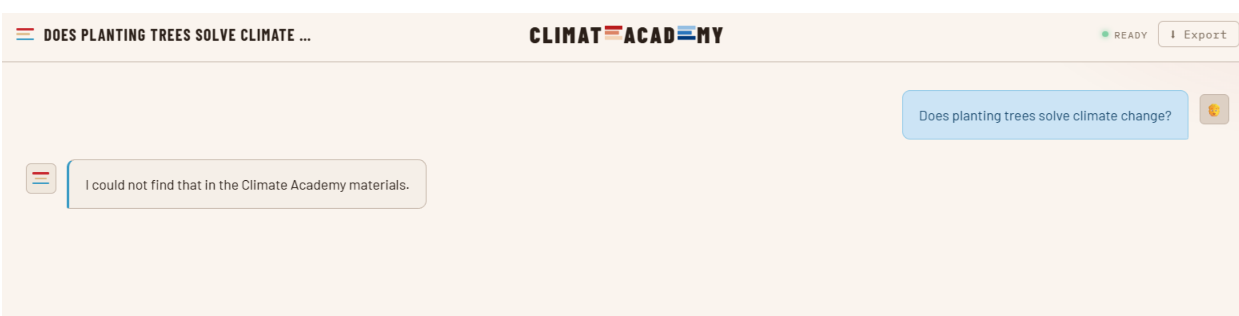


Figure 4: Testing the chatbot's handling of climate misconceptions

Overall, the QA process resembled human-centred evaluation of AI systems, where emphasis was placed not only on factual correctness but also on pedagogy, user experience, accessibility, and conversational effectiveness.

3.2 Creative Science Communication

Designed and produced educational video content based on the Climate Academy. Completed video: "Climate Anxiety: Turning Worry into Climate Action."

The video:

- Summarised the key concepts discussed in the Climate Academy chapter
- Explained climate anxiety as a rational emotional response
- Emphasised action and education as constructive responses
- Introduced the Climate Academy chatbot as an interactive learning platform

The content was designed for social media platforms using Canva and short-form storytelling principles.

Video URL: https://youtu.be/gEc02k7Qk1g

Figure 5: Thumbnail of the video for reference

Video title: Turning climate anxiety to climate action with #semanticClimate's Climate Academy chatbot

Description: Climate change and its repercussions can feel overwhelming, but understanding is often the first step towards meaningful action. Climate Academy chatbot could be your new interactive companion for climate-related learning and education, designed to empower students to make informed decisions and contribute to a sustainable future!

3.3 Content Ideation

Developed concepts for future educational videos inspired by the Climate Academy. The proposed strategy emphasised storytelling, literature, popular culture, visual analogies, and inquiry-based learning, rather than conventional educational summaries.

Examples included:

- Finding Nemo and coral reefs
- WALL-E and Spaceship Earth
- Jurassic Park and mass extinction
- Dominoes and climate tipping points

These concepts were documented for future development within #semanticClimate.

3.4 Project Meetings

Participated in daily team meetings, technical discussions, chatbot demonstrations, and collaborative brainstorming sessions.

These meetings provided insights into:

- AI workflow design
- Educational technology
- Software development
- Interdisciplinary collaboration

4. Skills Acquired

4.1 Technical Skills

Technical Skills
• Chatbot evaluation
• Prompt testing
• Science communication
• Educational content development
• Canva-based visual design
• Climate education resources
• GitHub orientation
• Markdown documentation
• Exposure to Python workflows

4.2 Professional Skills

Professional Skills
• Interdisciplinary collaboration
• Remote teamwork
• Scientific communication
• Project planning
• Creative problem solving
• Feedback integration

5. Key Learnings

This internship demonstrated how Artificial Intelligence can serve as a powerful educational tool when combined with accurate scientific information and thoughtful instructional design.

One of the most valuable insights was understanding that climate education extends beyond communicating scientific facts. Effective engagement requires storytelling, relatable examples, emotional connection, and opportunities for learners to actively explore concepts.

Working alongside researchers from diverse backgrounds highlighted the importance of integrating science, education, and technology to create meaningful public engagement with climate issues.

6. Reflections

One area I found particularly interesting was exploring ways of making climate education more engaging through short-form digital media.

Rather than simply summarising educational material, I began developing ideas that use literature, films, historical events, and everyday experiences as entry points into climate science. Such approaches could improve learner curiosity and naturally encourage continued interaction with AI-powered educational tools like CAbot.

This experience strengthened my interest in science communication, biodiversity informatics, and AI-assisted environmental education.

7. Suggestions for Future Creative Content

Potential directions for future content include:

- Climate Through Stories

- Climate Through Popular Culture
- Ask CAbot
- Climate Myths
- Climate Concepts in 60 Seconds
- Climate in Everyday Life

These approaches may help broaden audience engagement while directing learners towards deeper exploration through the chatbot.

8. Conclusion

The #semanticClimate internship provided valuable exposure to AI-assisted climate education, interdisciplinary collaboration, and digital science communication. It strengthened my understanding of how technological tools can improve climate literacy while highlighting the importance of accessible and engaging educational content.

The internship has significantly influenced my academic interests at the intersection of biodiversity informatics, environmental science, and science communication, and will serve as a strong foundation for future research and outreach activities.